

Assignment 3 - Differential Expression with edgeR (mobData)

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Part A - Setup and data

```
# Run once if needed:
# if (!require("BiocManager", quietly = TRUE)) install.packages("BiocManager")
# BiocManager::install(c("edgeR", "baySeq"))
library(edgeR)
library(baySeq)
```

```
# Relative path: works when this .Rmd sits in the repo root next to /data
load("data/mobData.RData")
head(mobData)
```

```
##      SL236 SL260 SL237 SL238 SL239 SL240
## [1,]    21    52     4     4    86    68
## [2,]    18    21     1     5     1     1
## [3,]     1     2     2     3     0     0
## [4,]    68    87   270   184   396   368
## [5,]    68    87   270   183   396   368
## [6,]     1     0     6    10     6    12
```

```
dim(mobData)
```

```
## [1] 3000    6
```

```
mobDataGroups <- c("MM", "MM", "WM", "WM", "WW", "WW")
# MM = triple mutant shoot grafted onto triple mutant root
# WM = wild-type shoot grafted onto triple mutant root
# WW = wild-type shoot grafted onto wild-type root
data(mobAnnotation)
head(mobAnnotation)
```

```
##   chr start  end
## 1   1   789  869
## 2   1  8641 8700
## 3   1 10578 10599
## 4   1 17041 17098
## 5   1 17275 17318
## 6   1 17481 17527
```

```
d <- DGEList(counts = mobData, group = factor(mobDataGroups))
d
```

```
## An object of class "DGEList"
## $counts
##   SL236 SL260 SL237 SL238 SL239 SL240
## 1    21    52     4     4    86    68
## 2    18    21     1     5     1     1
## 3     1     2     2     3     0     0
## 4    68    87   270   184   396   368
## 5    68    87   270   183   396   368
## 2995 more rows ...
##
## $samples
##      group lib.size norm.factors
## SL236    MM  152461             1
## SL260    MM  309995             1
## SL237    WM  216924             1
## SL238    WM  208841             1
## SL239    WW  258404             1
## SL240    WW  276434             1
```

Part B - Lab steps from “Filtering the data”

Filtering

```
dim(d)
```

```
## [1] 3000    6
```

```
d.full <- d                                # keep a copy of the unfiltered object
head(d$counts)
```

```
##   SL236 SL260 SL237 SL238 SL239 SL240
## 1    21    52     4     4    86    68
## 2    18    21     1     5     1     1
## 3     1     2     2     3     0     0
## 4    68    87   270   184   396   368
## 5    68    87   270   183   396   368
## 6     1     0     6    10     6    12
```

```
head(cpm(d))
```

```
##      SL236      SL260      SL237      SL238      SL239      SL240
## 1 137.740143 167.744641  18.439638  19.15333  332.812186  245.9900
## 2 118.062980  67.743028   4.609909  23.94166   3.869909   3.6175
## 3  6.559054   6.451717   9.219819  14.36500   0.000000   0.0000
## 4 446.015702 280.649688 1244.675555 881.05305 1532.484017 1331.2400
## 5 446.015702 280.649688 1244.675555 876.26472 1532.484017 1331.2400
## 6  6.559054   0.000000  27.659457  47.88332   23.219455   43.4100
```

```
apply(d$counts, 2, sum)           # total counts per sample
```

```
## SL236 SL260 SL237 SL238 SL239 SL240
## 152461 309995 216924 208841 258404 276434
```

```
# keep loci with > 100 counts-per-million in at least 2 samples
keep <- rowSums(cpm(d) > 100) >= 2
d <- d[keep, ]
dim(d)
```

```
## [1] 724 6
```

```
# reset library sizes after filtering
d$samples$lib.size <- colSums(d$counts)
d$samples
```

```
##      group lib.size norm.factors
## SL236    MM   145153           1
## SL260    MM   294928           1
## SL237    WM   207227           1
## SL238    WM   200563           1
## SL239    WW   242628           1
## SL240    WW   259990           1
```

Normalization (TMM)

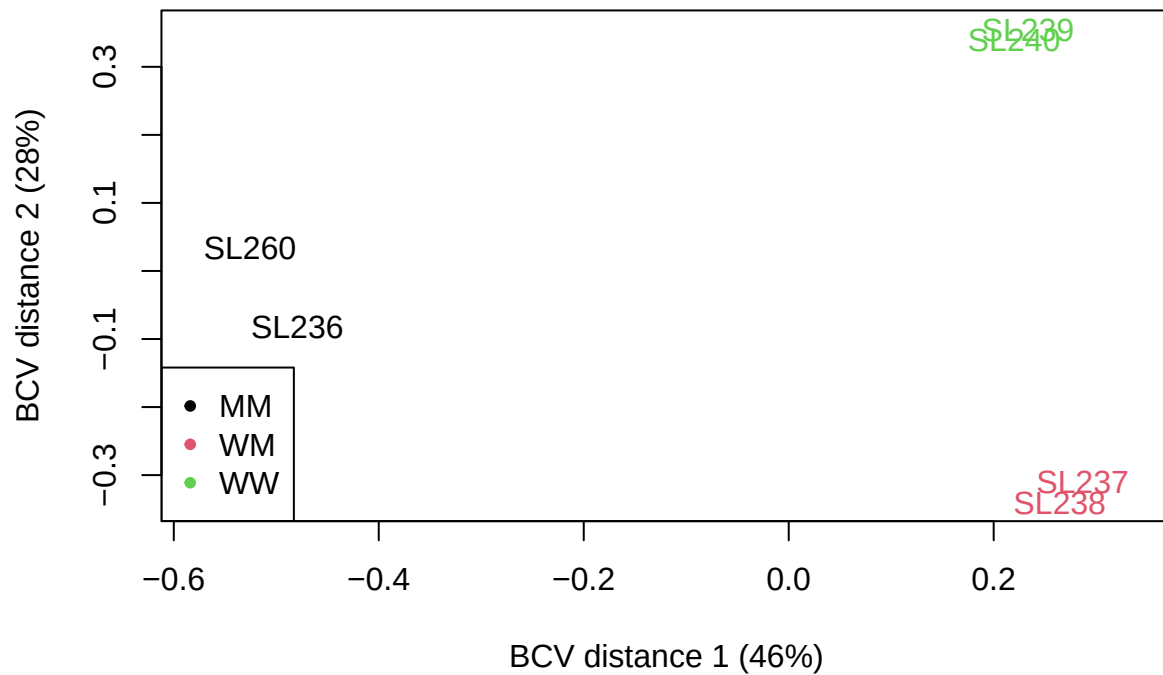
```
d <- calcNormFactors(d)
d$samples
```

```
##      group lib.size norm.factors
## SL236    MM   145153   1.0284561
## SL260    MM   294928   0.8947244
## SL237    WM   207227   1.0128068
## SL238    WM   200563   0.7695941
## SL239    WW   242628   1.1946131
## SL240    WW   259990   1.1671034
```

Data exploration (MDS)

```
plotMDS(d, method = "bcv", col = as.numeric(d$samples$group),
        main = "MDS plot (BCV distance)")
legend("bottomleft", legend = levels(d$samples$group), col = 1:3, pch = 20)
```

MDS plot (BCV distance)



Estimating dispersion - classic (qCML) model

```
d1 <- estimateCommonDisp(d, verbose = TRUE)
```

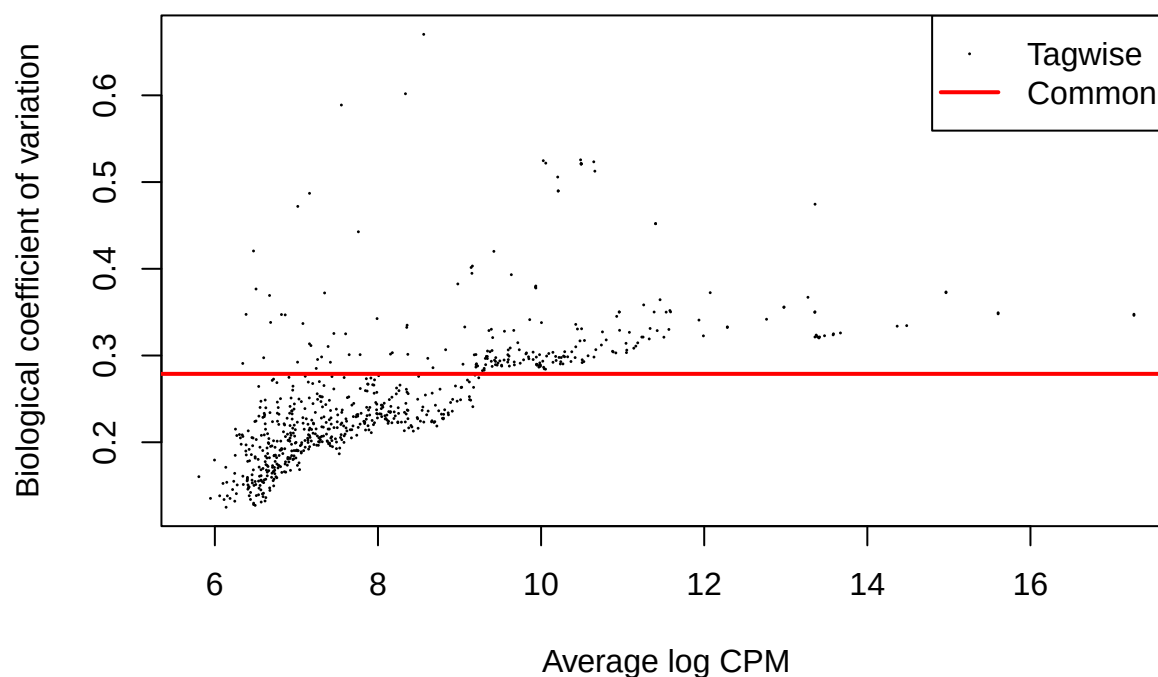
```
## Disp = 0.07776 , BCV = 0.2789
```

```
d1 <- estimateTagwiseDisp(d1)
names(d1)
```

```
## [1] "counts"          "samples"          "common.dispersion"
## [4] "pseudo.counts"   "pseudo.lib.size"  "AveLogCPM"
## [7] "prior.df"        "prior.n"          "tagwise.dispersion"
## [10] "span"
```

```
plotBCV(d1, main = "BCV vs average log CPM (classic model)")
```

BCV vs average log CPM (classic model)



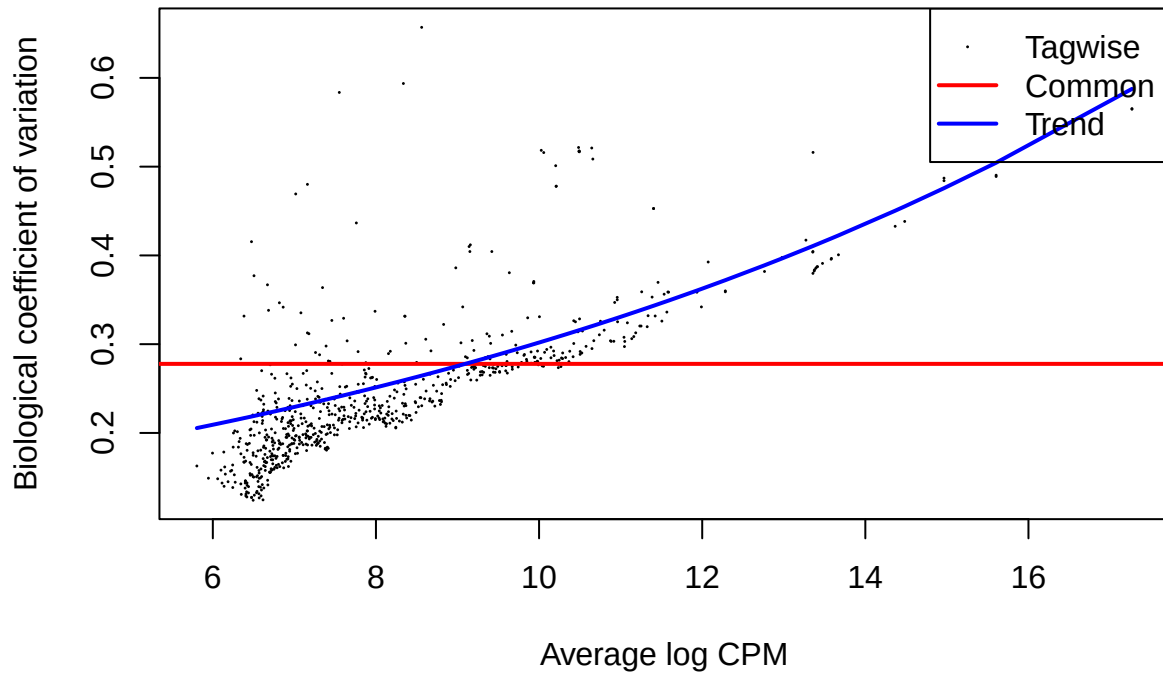
Estimating dispersion - GLM

```
design.mat <- model.matrix(~ 0 + d$samples$group)
colnames(design.mat) <- levels(d$samples$group)
design.mat
```

```
##      MM WM WW
## 1    1  0  0
## 2    1  0  0
## 3    0  1  0
## 4    0  1  0
## 5    0  0  1
## 6    0  0  1
## attr(,"assign")
## [1] 1 1 1
## attr(,"contrasts")
## attr(,"contrasts")$'d$samples$group'
## [1] "contr.treatment"
```

```
d2 <- estimateGLMCommonDisp(d, design.mat)
d2 <- estimateGLMTrendedDisp(d2, design.mat, method = "power")
d2 <- estimateGLMTagwiseDisp(d2, design.mat)
plotBCV(d2, main = "BCV vs average log CPM (GLM)")
```

BCV vs average log CPM (GLM)



(“Comparing the models in DESeq and edgeR” is skipped, as instructed.)

Differential expression - exact test (lab version)

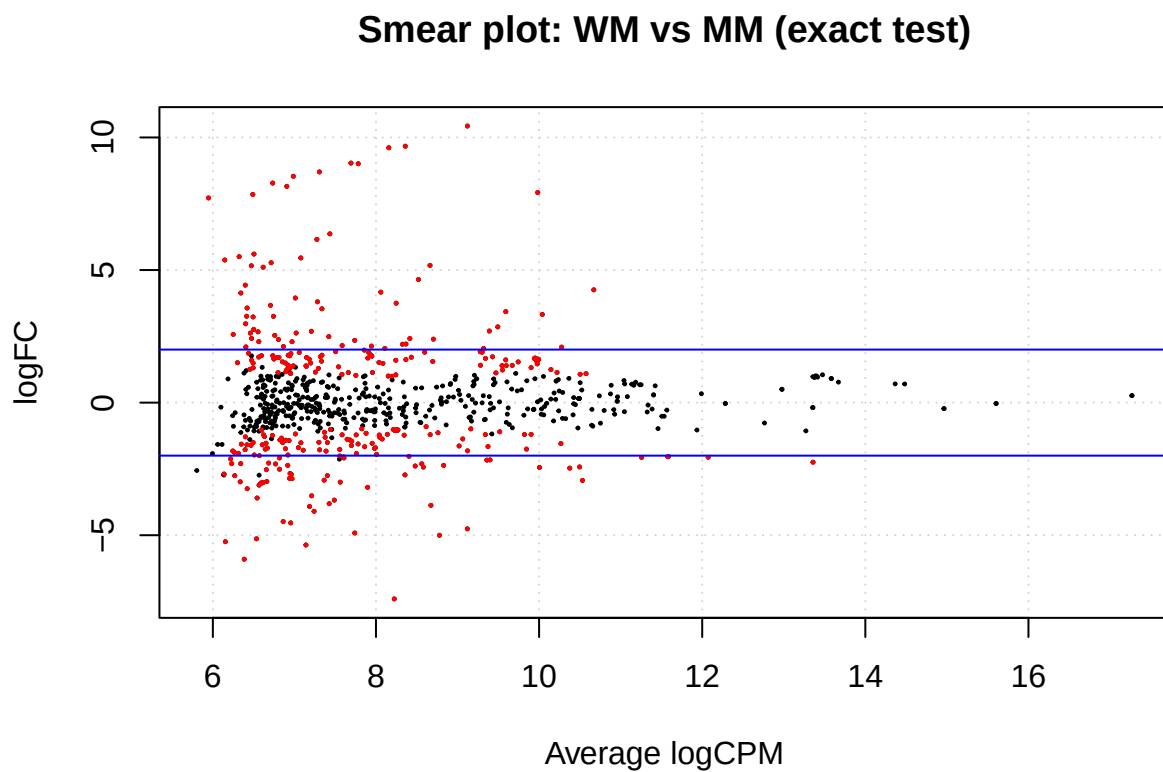
```
et12 <- exactTest(d1, pair = c(1, 2)) # WM vs MM
et13 <- exactTest(d1, pair = c(1, 3)) # WW vs MM
et23 <- exactTest(d1, pair = c(2, 3)) # WW vs WM
topTags(et12, n = 10)
```

```
## Comparison of groups: WM-MM
##      logFC  logCPM      PValue      FDR
## 74  10.430006 9.120844 2.695417e-29 1.951482e-26
## 1717 7.922931 9.983904 9.410276e-29 3.406520e-26
## 1111 9.670350 8.359530 1.063583e-24 2.566780e-22
## 1334 9.611448 8.157823 8.514229e-24 1.541075e-21
## 2322 5.174223 8.662531 2.945665e-23 4.265323e-21
## 625  -7.403650 8.224554 1.548105e-22 1.868047e-20
## 2537 -5.003070 8.778324 3.036411e-22 3.140516e-20
## 1353 9.032995 7.693571 7.824093e-21 7.080804e-19
## 1212 -4.759955 9.120041 2.824606e-20 2.272238e-18
## 2076 6.365397 7.434757 2.215394e-18 1.603945e-16
```

```
de1 <- decideTests(et12, adjust.method = "BH", p.value = 0.05)
summary(de1)
```

```
##          WM-MM
## Down      140
## NotSig    436
## Up        148
```

```
de1tags12 <- rownames(d1)[as.logical(de1)]
plotSmeaer(et12, de.tags = de1tags12, main = "Smeaer plot: WM vs MM (exact test)")
abline(h = c(-2, 2), col = "blue")
```



Differential expression - GLM (lab version)

```
fit <- glmFit(d2, design.mat)
lrt12 <- glmLRT(fit, contrast = c(1, -1, 0))
lrt13 <- glmLRT(fit, contrast = c(1, 0, -1))
lrt23 <- glmLRT(fit, contrast = c(0, 1, -1))
topTags(lrt12, n = 10)
```

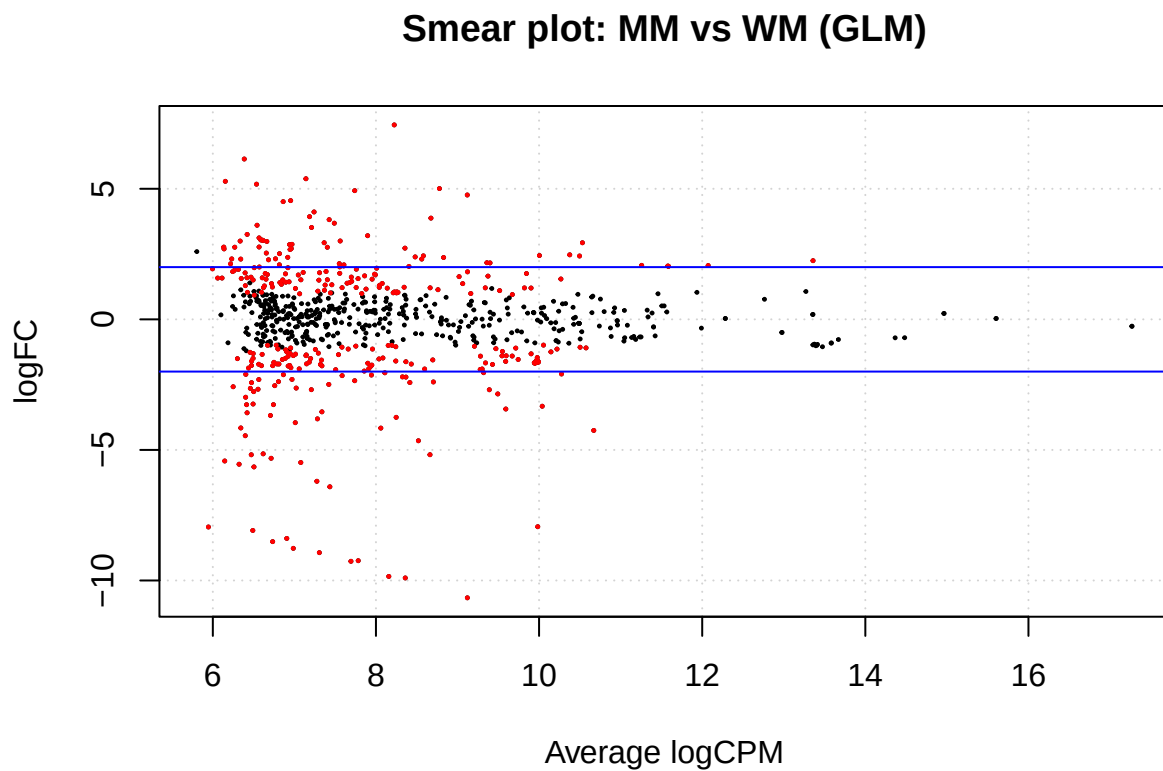
```
## Coefficient: 1*MM -1*WM
##          logFC  logCPM      LR      PValue      FDR
```

```
## 1717 -7.940313 9.983895 137.29913 1.037219e-31 7.509465e-29
## 74 -10.663209 9.120868 124.67026 6.009523e-29 2.175447e-26
## 1111 -9.903262 8.359566 116.49191 3.708887e-27 8.950781e-25
## 1334 -9.844242 8.157813 115.81279 5.223508e-27 9.454550e-25
## 625 7.445548 8.224665 104.94947 1.252903e-24 1.814203e-22
## 1353 -9.265860 7.693619 97.65123 4.989604e-23 6.020789e-21
## 2322 -5.182636 8.662524 94.82371 2.081174e-22 2.152529e-20
## 2537 5.008245 8.778375 94.10169 2.997229e-22 2.712492e-20
## 1696 -8.933420 7.306297 85.07723 2.869353e-20 2.308235e-18
## 764 -9.242198 7.782935 84.24614 4.368566e-20 3.162842e-18
```

```
de2 <- decideTests(lrt12, adjust.method = "BH", p.value = 0.05)
summary(de2)
```

```
##          1*MM -1*WM
## Down           156
## NotSig         416
## Up             152
```

```
de2tags12 <- rownames(d2)[as.logical(de2)]
plotSmear(lrt12, de.tags = de2tags12, main = "Smear plot: MM vs WM (GLM)")
abline(h = c(-2, 2), col = "blue")
```



Part C - Analyses for the homework questions

All WW vs MM comparisons below are oriented the same way: $\log FC = \log_2(WW / MM)$, so “down-regulated” means lower expression in WW than in MM.

```
# Q3: exact test, COMMON dispersion, WW vs MM
et_WW_MM <- exactTest(d1, pair = c("MM", "WW"), dispersion = "common")
de_et <- decideTests(et_WW_MM, adjust.method = "BH", p.value = 0.05)
summary(de_et)
```

```
##          WW-MM
## Down      114
## NotSig    452
## Up        158
```

```
topTags(et_WW_MM, n = 10)
```

```
## Comparison of groups: WW-MM
##          logFC  logCPM      PValue      FDR
## 1717  7.141029 9.983904 7.463051e-26 4.019025e-23
## 74    10.565467 9.120844 1.110228e-25 4.019025e-23
## 490   6.127494 8.541482 2.415851e-23 5.830254e-21
## 1111  9.764010 8.359530 2.782106e-20 5.035612e-18
## 1334  9.369450 8.157823 8.424120e-18 1.155618e-15
## 1963 -6.434608 6.954298 9.576948e-18 1.155618e-15
## 764   9.231516 7.782850 6.190143e-17 6.402376e-15
## 1528  5.063371 8.520379 1.314822e-16 1.189914e-14
## 1212 -3.957256 9.120041 2.971863e-16 2.390699e-14
## 2500  4.107649 9.592264 3.344310e-16 2.421280e-14
```

```
n_down_et <- sum(de_et == -1)
n_up_et   <- sum(de_et == 1)
n_ns_et   <- sum(de_et == 0)
```

```
# Q4: GLM likelihood-ratio test, WW vs MM (contrast WW - MM)
lrt_WW_MM <- glmLRT(fit, contrast = c(-1, 0, 1))
de_glm <- decideTests(lrt_WW_MM, adjust.method = "BH", p.value = 0.05)
summary(de_glm)
```

```
##          -1*MM 1*WW
## Down           123
## NotSig         414
## Up             187
```

```
topTags(lrt_WW_MM, n = 10)
```

```
## Coefficient: -1*MM 1*WW
##          logFC  logCPM      LR      PValue      FDR
## 490   6.116142 8.541820 122.88815 1.475358e-28 1.068159e-25
## 74    10.426218 9.120868 117.53796 2.188704e-27 7.923109e-25
```

```
## 1717  7.131777 9.983895 111.35563 4.945390e-26 1.193487e-23
## 1111  9.625056 8.359566 107.37515 3.683976e-25 6.667996e-23
## 1027 -5.257532 6.421199  97.56969 5.199371e-23 7.528689e-21
## 1334  9.230074 8.157813  94.06834 3.048153e-22 3.678105e-20
## 1528  5.069486 8.520478  91.99614 8.684576e-22 8.982333e-20
## 1963 -6.420110 6.954254  89.57115 2.958063e-21 2.677047e-19
## 1353  8.873376 7.693619  85.26993 2.602918e-20 2.093903e-18
## 764   9.092243 7.782935  81.16124 2.080327e-19 1.506157e-17
```

```
n_down_glm <- sum(de_glm == -1)
n_up_glm   <- sum(de_glm ==  1)

down_et    <- rownames(d1)[de_et == -1]
down_glm   <- rownames(d2)[de_glm == -1]
up_et      <- rownames(d1)[de_et ==  1]
up_glm     <- rownames(d2)[de_glm ==  1]
n_down_both <- length(intersect(down_et, down_glm))
n_up_both   <- length(intersect(up_et, up_glm))
n_de_et     <- n_down_et + n_up_et
n_de_both   <- n_down_both + n_up_both
pct_shared  <- if (n_de_et > 0) round(100 * n_de_both / n_de_et) else NA

comparison <- data.frame(
  Method = c("Exact test (common dispersion)", "GLM LRT (tagwise dispersion)", "Shared by both"),
  Down   = c(n_down_et, n_down_glm, n_down_both),
  Up     = c(n_up_et, n_up_glm, n_up_both))
knitr::kable(comparison, caption = "WW vs MM, BH-adjusted p < 0.05")
```

Table 1: WW vs MM, BH-adjusted $p < 0.05$

Method	Down	Up
Exact test (common dispersion)	114	158
GLM LRT (tagwise dispersion)	123	187
Shared by both	101	156

Note: the lab's lrt13 uses contrast $c(1,0,-1) = MM - WW$, so its "Down" and "Up" rows are swapped relative to this table. The genes are the same.

```
# Q5: a different edgeR method - quasi-likelihood F-test (glmQLFit / glmQLFTest)
qlfit <- glmQLFit(d2, design.mat)
qlf_WW_MM <- glmQLFTest(qlfit, contrast = c(-1, 0, 1))
de_ql <- decideTests(qlf_WW_MM, adjust.method = "BH", p.value = 0.05)
summary(de_ql)
```

```
##          -1*MM 1*WW
## Down           103
## NotSig         436
## Up             185
```

```
n_down_q1 <- sum(de_q1 == -1)
n_up_q1   <- sum(de_q1 == 1)
```

Homework answers

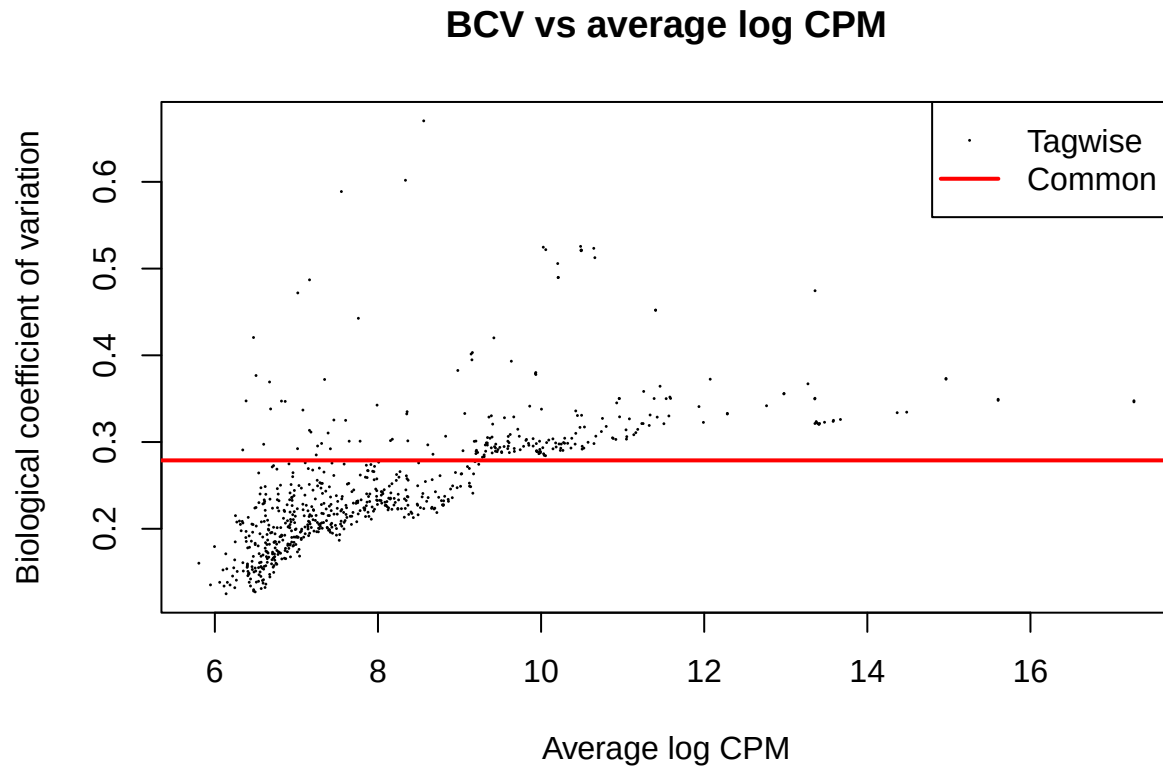
Question 1 - What are SL239 and SL240?

```
data.frame(Sample = colnames(d), Group = d$samples$group,
           lib.size = d$samples$lib.size)
```

```
##   Sample Group lib.size
## 1  SL236    MM  145153
## 2  SL260    MM  294928
## 3  SL237    WM  207227
## 4  SL238    WM  200563
## 5  SL239    WW  242628
## 6  SL240    WW  259990
```

SL239 and SL240 are the two biological replicate libraries of the **WW** group: small-RNA sequencing samples from *Arabidopsis* plants in which a **wild-type shoot was grafted onto a wild-type root**. They are the control (fully wild-type) condition of the grafting experiment, against which the MM samples (SL236, SL260; triple-mutant shoot on triple-mutant root) and the WM samples (SL237, SL238; wild-type shoot on triple-mutant root) are compared. Each column holds the counts for the 3,000 small-RNA loci in that sample.

Question 2 - Describe the CPM vs BCV plot



The plot shows how biological variability relates to abundance for the 724 loci kept after filtering. The x-axis is average expression (log2 counts per million) and the y-axis is the biological coefficient of variation (BCV), the square root of the negative-binomial dispersion. BCV is the relative variation between biological replicates. The red line is the common BCV of 0.279, meaning that a typical locus varies by about 28% between replicates. The black points are the tagwise (gene-specific) BCVs.

BCV increases as abundance rises (Spearman $\rho = 0.8$; median BCV 0.174 in the lowest-abundance quartile vs 0.307 in the highest). The spread of points is widest at low CPM, where small counts make estimates noisy. Some loci lie well above the common line, so variability differs from locus to locus. This supports filtering out low-count loci and using tagwise (or trended) dispersion rather than one common value for every locus.

(Check this paragraph against your rendered figure and edit any wording that doesn't match.)

Question 3 - Downregulated genes, WW vs MM, exact test with common dispersion

Using `exactTest(d1, pair = c("MM","WW"), dispersion = "common")` and a Benjamini-Hochberg FDR cutoff of 0.05, **114 loci are downregulated** in WW relative to MM. Another 158 are upregulated, and 452 are not significant.

Question 4 - Comparison with the GLM

Table 2: WW vs MM, BH-adjusted $p < 0.05$

Method	Down	Up
Exact test (common dispersion)	114	158
GLM LRT (tagwise dispersion)	123	187
Shared by both	101	156

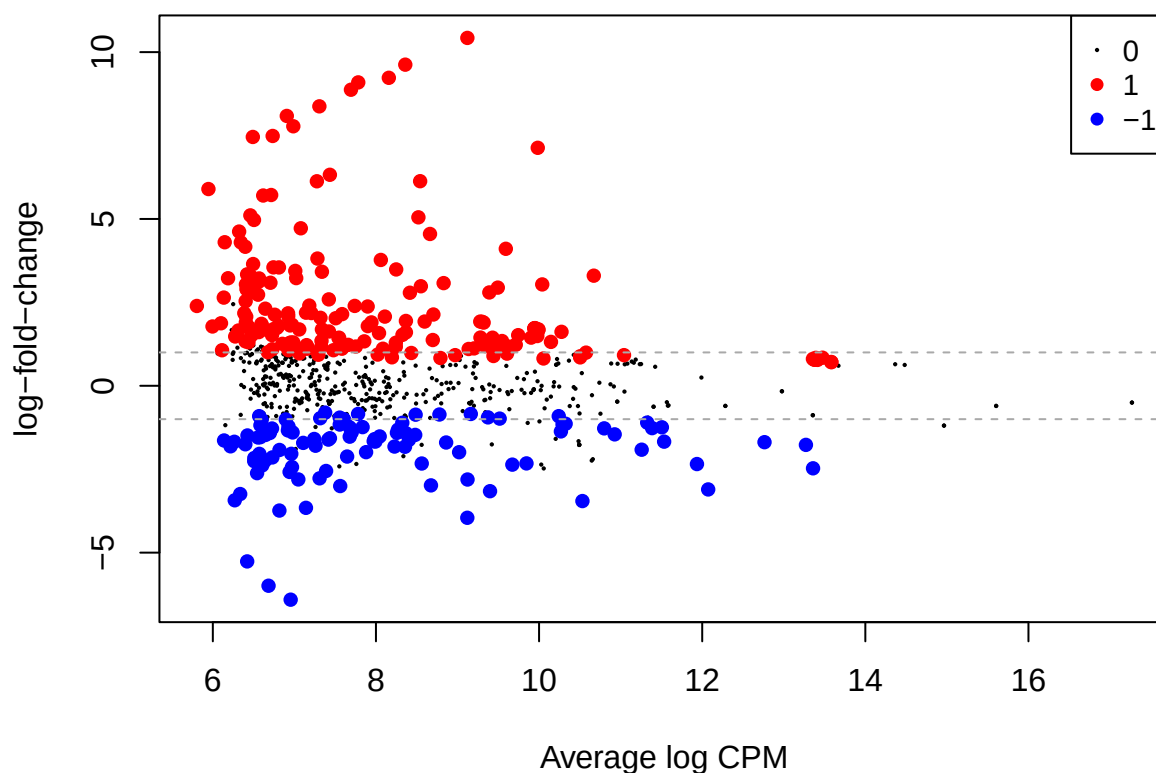
The GLM likelihood-ratio test (`glmFit` + `glmLRT`, contrast WW - MM) finds 123 downregulated and 187 upregulated loci, compared with 114 and 158 from the exact test. 257 loci (94% of the exact-test calls) are significant in the same direction in both. The counts differ mainly because of how variability is modelled. The exact test applies one common dispersion to every locus, whereas the GLM uses tagwise dispersions estimated with the full design matrix. Loci that are more variable than average therefore lose significance under the GLM, and unusually stable loci gain it. The GLM is also the more flexible framework: it can handle multiple factors, batch effects and arbitrary contrasts. The exact test only works for simple two-group comparisons.

(If you used the lab's `lrt13`, contrast $c(1,0,-1) = MM - WW$, your “up” and “down” counts will be swapped. State the direction you used.)

Question 5 - New plot with a different edgeR method

For this plot I used edgeR's **quasi-likelihood (QL) F-test** (`glmQLFit` + `glmQLFTest`), which the edgeR User's Guide recommends for bulk RNA-seq. It is different from the exact test and the likelihood-ratio test used in the lab. The plot is an **MD (mean-difference) plot** of the WW vs MM comparison.

MD plot: WW vs MM (quasi-likelihood F-test, FDR < 0.05)



Each point is a locus. The x-axis is its average abundance (log2 CPM), and the y-axis is its log2 fold change (WW / MM). Red points are significantly upregulated in WW ($n = 185$), blue points are significantly downregulated ($n = 103$), and black points are not significant. The dashed lines mark a 2-fold change.

The plot shows that most loci are centered on $\log_{2}FC = 0$, which confirms that TMM normalization worked and that most small-RNA loci do not change between genotypes. The significant loci are spread across the whole range of abundance, so the calls aren't driven only by highly expressed loci. At low abundance, larger fold changes are needed to reach significance because count noise is higher. The QL method also accounts for uncertainty in the dispersion estimates, so it is usually a little more conservative than the LRT. That is why it calls 288 loci here, compared with 310 for the GLM LRT. The loci that remain significant under QL are the most reliable candidates for small RNAs whose abundance depends on the wild-type versus triple-mutant genotype.

```
sessionInfo()
```

```
## R version 4.6.1 (2026-06-24)
## Platform: x86_64-linux-gnu
## Running under: Ubuntu 26.04.1 LTS
##
## Matrix products: default
## BLAS: /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
## LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblas-p0.3.32.so; LAPACK version 3.12.0
##
```

```

## locale:
## [1] LC_CTYPE=en_US.UTF-8      LC_NUMERIC=C
## [3] LC_TIME=en_US.UTF-8      LC_COLLATE=en_US.UTF-8
## [5] LC_MONETARY=en_US.UTF-8  LC_MESSAGES=en_US.UTF-8
## [7] LC_PAPER=en_US.UTF-8     LC_NAME=C
## [9] LC_ADDRESS=C             LC_TELEPHONE=C
## [11] LC_MEASUREMENT=en_US.UTF-8 LC_IDENTIFICATION=C
##
## time zone: America/Toronto
## tzcode source: system (glibc)
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods    base
##
## other attached packages:
## [1] baySeq_2.46.0 edgeR_4.10.5 limma_3.68.5
##
## loaded via a namespace (and not attached):
## [1] cli_3.6.6      knitr_1.52      rlang_1.3.0
## [4] xfun_0.61      ote1_0.2.0      generics_0.1.4
## [7] S4Vectors_0.50.3 statmod_1.5.2    htmltools_0.5.9
## [10] stats4_4.6.1   locfit_1.5-9.12 rmarkdown_2.32
## [13] grid_4.6.1     Seqinfo_1.2.0   evaluate_1.0.5
## [16] abind_1.4-8     fastmap_1.2.0   yaml_2.3.12
## [19] IRanges_2.46.0 compiler_4.6.1   lattice_0.23-1
## [22] digest_0.6.39  parallel_4.6.1  GenomicRanges_1.64.0
## [25] tools_4.6.1    BiocGenerics_0.58.1

```